

Hawkeye EQMS · Training Records · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Tenant under test	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Training Records
Pharma purpose	Track assigned, in-progress and completed training for every employee. Auto-assign 'Read and Understood' records when SOPs publish. Capture competency assessment + knowledge-check results for audit readiness.
Compliance basis	ISO 9001:2015 §7.2 (competence) · 21 CFR 211.25 (personnel qualification) · 21 CFR 211.100 (training on SOPs) · EU GMP Chapter 2
Hawkeye routes	/training (frontend) · /api/training-records (backend)
Hawkeye model file	backend/src/models/TrainingRecordModel.js
Tenant module flag	modules.TRAINING

Records move ASSIGNED → IN_PROGRESS → COMPLETED (or OVERDUE / WAIVED / FAILED). The Wave 2 Training Auto-Assign agent fires when an SOP with requiresTrainingOnUpdate=true publishes — it identifies affected roles + creates a record per person + optionally drafts a knowledge-check question.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Records cover all 21 CFR 211.25 training types (onboarding / SOP read-and-understand / regulatory / GMP / safety / technical / process / quality system) 21 CFR 211.25	trainingType enum with 9 values matching CFR 211.25.	MET	
Auto-assign on SOP publish 21 CFR 211.100	Wave 2 Training Auto-Assign agent fires from /publish hook when requiresTrainingOnUpdate=true.	MET	
Competency level recorded (Aware / Competent / Proficient / Expert) ISO 9001 §7.2	competencyLevel enum on /complete.	MET	
Assessment captured (type / score / passed) 21 CFR 211.25(a)	assessment subdocument with assessmentType (WRITTEN_TEST/PRACTICAL/OBSERVATION/SIGN_OFF/ORAL_EXAM) + score + passed.	MET	
Due date + grace period tracking 21 CFR 211.25	dueDate field; auto-assign uses gracePeriodDays.	MET	
OVERDUE escalation when dueDate passes ISO 9001 §7.2(d)	Vercel cron (03:00 UTC daily) calls /api/quality/scan-overdue → flips records past dueDate to OVERDUE + writes notification-outbox rows.	MET	
AI-drafted knowledge-check question per SOP rev —	Auto-assign agent optionally drafts an MCQ via Free Gemini.	MET	
Waiver flow with reason + waivedBy + waivedAt 21 CFR 211.25(c)	WAIVED status + waiverReason + waivedBy fields.	MET	
21 CFR Part 11 e-signature on training completion 21 CFR Part 11	Generic e-sig endpoint exists; not enforced on /complete today.	PARTIAL	Wire /api/electronic-signatures/sign on /complete

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Rebecca Kim Training Coordinator (admin)	training.coord@novex-pharma.demo	Assigns training, monitors compliance, processes waivers.	ASSIGNED WAIVED FAILED
Trainees (any user) varies	all 11 personas can be trainees	Complete assigned trainings + assessment.	IN_PROGRESS COMPLETED
Sarah O'Brien (Doc Control) admin · upstream	doc.control@novex-pharma.demo	Publishes SOPs that fire auto-assign.	ASSIGNED (downstream of SOP publish)

4. Feature catalogue (click-by-click)

4.1 Training register

What it does	Lists all training records with status filter + assignedTo + dueDate.
Menu / route	/training
Available to roles	any tenant viewer
Backend API	GET /api/training-records

CLICK-BY-CLICK

- NAVIGATE** Click 'Training' in the top nav
→ Page renders
- WAIT** Spinner clears
→ Seeded rows visible: 2 COMPLETED for Kenji, 1 ASSIGNED for production head

Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Admin

Training

Training & Competency

Training assignments, completion tracking, and competency records

Completion Rate

50%

8 total

Status

Environmental monitoring · Grade C

SOP_READ_AND_UNDERSTAND · Due 10/19/2025 · Completed 10/9/2025

COMPLETED

Dissolution testing · USP <711> · rev 7.2

SOP_READ_AND_UNDERSTAND · Due 12/14/2025 · Completed 12/2/2025

COMPLETED

Blend-time verification · Line 2 · rev 5.0

SOP_READ_AND_UNDERSTAND · Due 5/7/2026

ASSIGNED

E2E Lifecycle · dissolution method read-and-understood

SOP_READ_AND_UNDERSTAND · Due 5/8/2026

IN_PROGRESS

E2E Lifecycle · dissolution method read-and-understood

SOP_READ_AND_UNDERSTAND · Due 5/8/2026 · Completed 4/24/2026

Competency: COMPETENT

COMPLETED

E2E Lifecycle · dissolution method read-and-understood

SOP_READ_AND_UNDERSTAND · Due 5/8/2026 · Completed 4/24/2026

Competency: COMPETENT

COMPLETED

p

SOP_READ_AND_UNDERSTAND · Due 5/14/2026

ASSIGNED

probe

SOP_READ_AND_UNDERSTAND · Due 5/14/2026

ASSIGNED

state-screens/training-list.png

4.2 + Assign Training dialog

What it does	Manually assign a training to a user.
Menu / route	/training · top-right '+ Assign Training' button
Available to roles	admin · trainer
Backend API	POST /api/training-records

CLICK-BY-CLICK

- CLICK** Click '+ Assign Training'
→ Dialog opens
- CLICK** Pick trainee from user dropdown
→ required
- TYPE** Fill trainingCode (e.g. SOP-QC-014@v7.2)
→ required
- TYPE** Fill trainingTitle
→ required
- CLICK** Pick trainingType (SOP_READ_AND_UNDERSTAND / GMP / SAFETY / etc.)
→ required
- CLICK** Pick dueDate
→ required
- CLICK** Click 'Save'
→ Row appears with status=ASSIGNED

FIELDS CAPTURED

Field	Required	Validation / values	Notes
traineeId	Yes	ObjectId · user	
trainingCode	Yes	string e.g. SOP-XXX@vN	
trainingTitle	Yes	string	
trainingType	Yes	ONBOARDING SOP_READ_AND_UNDERSTAND REGULATORY GMP SAFETY TECHNICAL PROCESS QUALITY_SYSTEM CUSTOM	
dueDate	Yes	ISO date	

4.3 Start training (ASSIGNED → IN_PROGRESS)

What it does	Trainee marks the record as started.
Menu / route	Training row · 'Start' button
Available to roles	trainee (own record)
Backend API	PUT /api/training-records/:id (status=IN_PROGRESS)

CLICK-BY-CLICK

- CLICK** Click 'Start' on your assigned row
→ Status flips to IN_PROGRESS

4.4 Complete (with assessment)

What it does	Captures competencyLevel + assessment + duration + (optional) e-signature.
Menu / route	Training row · 'Mark Complete' button
Available to roles	trainee
Backend API	POST /api/training-records/:id/complete

CLICK-BY-CLICK

- CLICK** Click 'Mark Complete'
→ Drawer with assessment form
- CLICK** Pick competencyLevel (AWARE / COMPETENT / PROFICIENT / EXPERT)
→ required
- CLICK** Pick assessment.type (WRITTEN_TEST / PRACTICAL / OBSERVATION / SIGN_OFF / ORAL_EXAM)
→ required
- TYPE** Fill assessment.score + tick passed
→ required
- TYPE** (Optional) Fill notes + trainingDurationMinutes
- CLICK** Click 'Submit'
→ Status=COMPLETED · completedAt set

4.5 Waive

What it does	Trainer waives a training with reason (e.g. trainee already qualified elsewhere).
Menu / route	Training row · 'Waive' button
Available to roles	trainer · admin
Backend API	PUT /api/training-records/:id (status=WAIVED, waiverReason, waivedBy, waivedAt)

CLICK-BY-CLICK

- 1. **CLICK** Click 'Waive'
→ Drawer with waiverReason field
- 2. **TYPE** Fill waiverReason
→ required
- 3. **CLICK** Click 'Confirm waiver'
→ Status=WAIVED · terminal

4.6 AI · Training auto-assign on SOP rev (Wave 2)

What it does	Fires automatically from doc-control /publish hook. Identifies affected roles + creates training-records + drafts knowledge-check.
Menu / route	Auto-fired (no manual button); records appear in /training
Available to roles	system (fired from SOP publish hook)
Backend API	POST /api/ai/training/auto-assign
AI assistance	AI Free Gemini for knowledge-check draft; rule engine for role mapping

CLICK-BY-CLICK

- 1. **WAIT** After an SOP with requiresTrainingOnUpdate=true publishes, wait ~5s
→ training-records inserted for users in affectedRoles + (optional) knowledgeCheck question on each record

5. Full lifecycle walkthrough — one transaction, end-to-end

One training record assigned, completed with assessment.

#	Persona	State transition	Action	API	Observed	Result
1	Rebecca Training Coordinator	– → ASSIGNED	Click '+ Assign Training' → trainee=Rebecca, code=E2E-LC, type=SOP_READ_AND_UNDERSTAND, due=+14d → Save	POST /api/training-records	Row visible with status=ASSIGNED	PASS

Expected DB state after this step:

```
training-records { _id, traineeId, trainingCode, trainingTitle, trainingType: 'SOP_READ_AND_UNDERSTAND', dueDate, status: 'ASSIGNED', assignedByUserId: rebecca._id }
```


AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Training Auto-Assign Agent (Wave 2)	ASSIGNED (creation)	POST /api/ai/training/auto-assign	Auto-fired from doc-control /publish hook	Identifies affected roles + creates training-records + (optional) drafts a knowledge-check MCQ	Free Gemini for question; rule engine for role mapping
(roadmap) AI competency gap analyzer	COMPLETED	(future)	Training Compliance dashboard	Compares assessment scores across roles + flags individuals + topics for follow-up training	Free Gemini

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
ASSIGNED	21 CFR 211.25(a) ISO 9001 §7.2(b)	trainee + trainingCode + trainingType + dueDate + assignedBy	training-records
COMPLETED	21 CFR 211.25(a) 21 CFR Part 11 §11.50	competencyLevel + assessment {type, score, passed} + completedAt + (e-sig when wired)	training-records + (electronic-signatures)
WAIVED	21 CFR 211.25(c)	waiverReason + waivedBy + waivedAt	training-records

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · training	ASSIGNED → IN_PROGRESS → COMPLETED with assessment	PASS	4/4 PASS · eqms-test-results-v2.pdf
eqms-cross-module.spec.ts · F2	Doc publish → AI auto-assign fires	PASS	F2 step 3 PASS

9. Known gaps + roadmap

- OVERDUE scheduler + notification — Cron that flips status to OVERDUE when dueDate passes + emails trainee + manager. **HIGH**
- Mandatory e-sig on /complete — Wire /api/electronic-signatures/sign for trainee + (optional) supervisor sign-off. **HIGH**
- Training matrix view (role × required SOPs) — Compliance grid per role. **MEDIUM**
- AI competency gap analyzer (Wave 3 roadmap) — Compare scores across roles + recommend remedial training. **MEDIUM**